

**2.11**

Logarithmic Functions

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: For base greater than 1, log is increasing and concave down.

2. Write one precise sentence explaining this principle: For base between 0 and 1, log is decreasing and concave up.

3. Write one precise sentence explaining this principle: The parent domain is $(0, \infty)$, range all real numbers, vertical asymptote $x = 0$.

4. Write one precise sentence explaining this principle: Transformations shift the vertical asymptote and change the domain.

Core

5. Analyze $f(x) = 2 \log_3(x + 4) - 1$.
6. If x is multiplied by 9, how much does $\log_3 x$ change?
7. Why can a logarithmic function have any real output?
8. State what the formula $f(x) = a \log_b(x - h) + k$ tells you and name any required restriction.
9. State what the formula vertical asymptote $x = h$ tells you and name any required restriction.
10. State what the formula $\log_b(cx) - \log_b x = \log_b c$ tells you and name any required restriction.

Medium

11. Analyze $\log_2 x$: domain, range, asymptote, monotonicity, concavity.
12. Analyze $\log_{1/3} x$: domain, range, asymptote, monotonicity, concavity.
13. Analyze $2 \ln(x - 4) + 1$: domain, range, asymptote, monotonicity, concavity.
14. Analyze $-\log_5(x + 2) + 3$: domain, range, asymptote, monotonicity, concavity.

15. Analyze $\ln(7 - x)$: domain, range, asymptote, monotonicity, concavity.

Hard

16. Find x-intercept of $3 \log_2(x - 1) - 6$.

17. If x is multiplied by 8, how does $\log_2 x$ change?

18. Construct a log function with VA $x=3$ and point $(4,-2)$, using base 10 and no vertical scale.

19. Compare $\log_2 x$ and $\log_4 x$.

Difficult

20. Find parameters a, h, k for $f = a \ln(x - h) + k$ with VA $x=2$ and points $(3,1), (e+2,5)$.

21. Determine all x where $\log_2 x = \log_4(x + 6)$.

22. Explain concavity of $\log_b x$ for $b > 1$ and $0 < b < 1$ without calculus language.

Challenge / HOT

23. Prove any two logarithmic functions with different bases are vertical scalings.

24. Construct a logarithmic function with domain $x < 5$, increasing, and vertical asymptote $x = 5$.

AP-Style Multiple Choice

25. Parent log domain is

- (A) $\mathbb{R}$
- (B) $(0, \infty)$
- (C) $[0, \infty)$
- (D) $(-\infty, 0)$

26. Parent log range is
(A) $\mathbb{R}$
(B) $(0, \infty)$
(C) $[0, 1]$
(D) $x > 0$
27. VA of $\ln(x + 7)$ is
(A) $x=7$
(B) $x=-7$
(C) $y=-7$
(D) $y=0$
28. For $b>1$, $\log$ is
(A) increasing
(B) decreasing
(C) constant
(D) even
29. $\log_{1/2} x$ is
(A) increasing
(B) decreasing
(C) not real
(D) linear
30. Multiplying input by b changes $\log_b x$ by
(A) b
(B) 1
(C) x
(D) 0

AP-Style Free Response

31. Analyze $f(x) = -3\log_2(4 - x) + 2$: domain, range, VA, monotonicity, concavity, intercept if exact.

32. A function changes by 6 whenever input is multiplied by 5 and satisfies $f(1)=2$. Assume logarithmic form. Construct f and find $f(125)$.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.